WT EXPERIMENT - 05

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Aim:

To simulate a project using Wokwi, an online electronics simulator, focusing on an detecting vehicle collisions using motion sensors and providing real-time location data to facilitate emergency response.

Theory:

Wokwi is a cloud-based platform designed for simulating electronics projects, particularly those involving microcontrollers like Arduino, ESP32, and STM32. It offers a user-friendly interface for designing and testing circuits without the need for physical hardware, making it ideal for beginners and experienced makers alike.

Overview of Wokwi

- Type of Site: Electronics simulation and design
- Owner: Wokwi
- URL: www.wokwi.com
- Commercial: Offers free and paid plans
- Registration: Required for full features
- Key Features:
    - Simulation: Supports real-time simulation of Arduino and other microcontroller projects.
    - Components: Includes a wide range of components such as LEDs, sensors, and actuators.
    - Collaboration: Allows users to share and collaborate on projects in real-time.
    - Debugging: Offers advanced debugging tools like GDB for Arduino and Raspberry Pi Pico.

Code:

The following code is for an RFID-based access control system using an Arduino board, a servo barrier, an LCD display, and a buzzer.

cpp

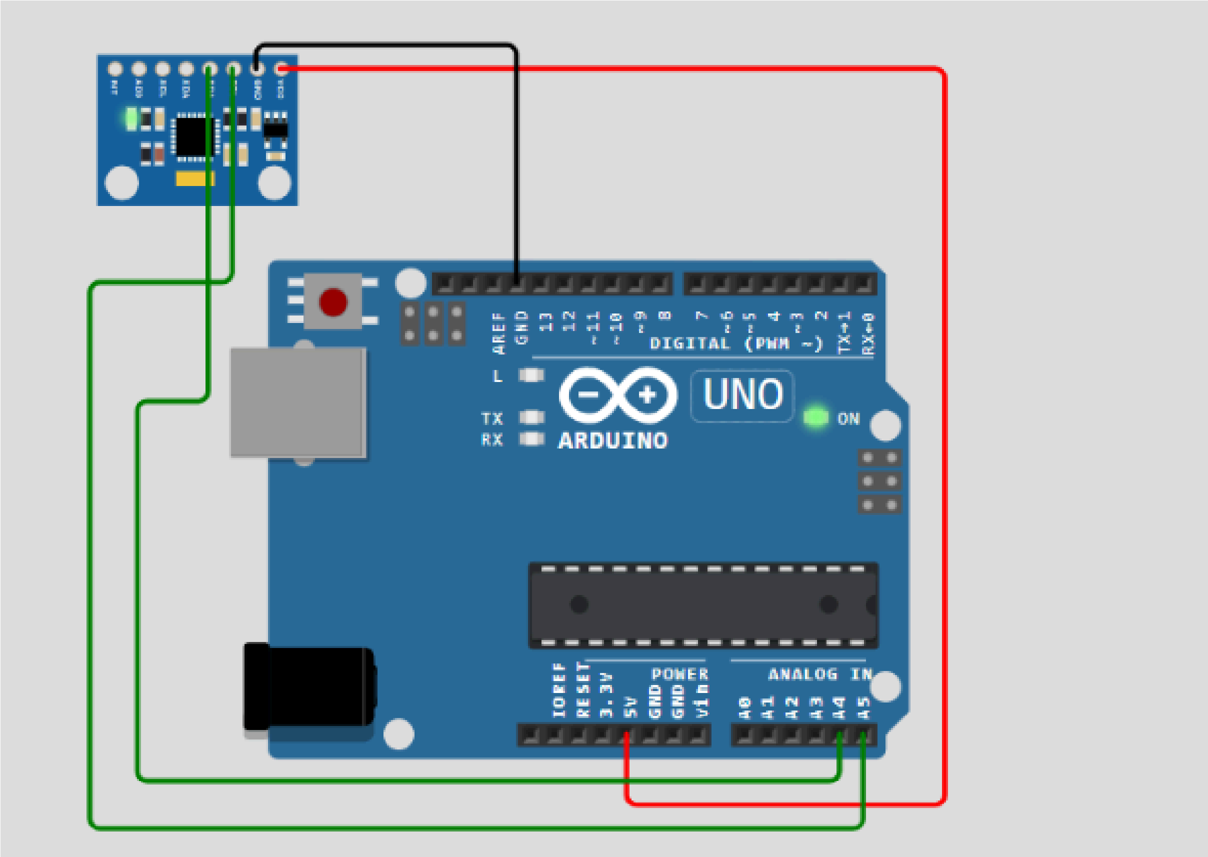
```
#include <Wire.h>
#include <MPU6050.h>

MPU6050 mpu;

void setup() {
```

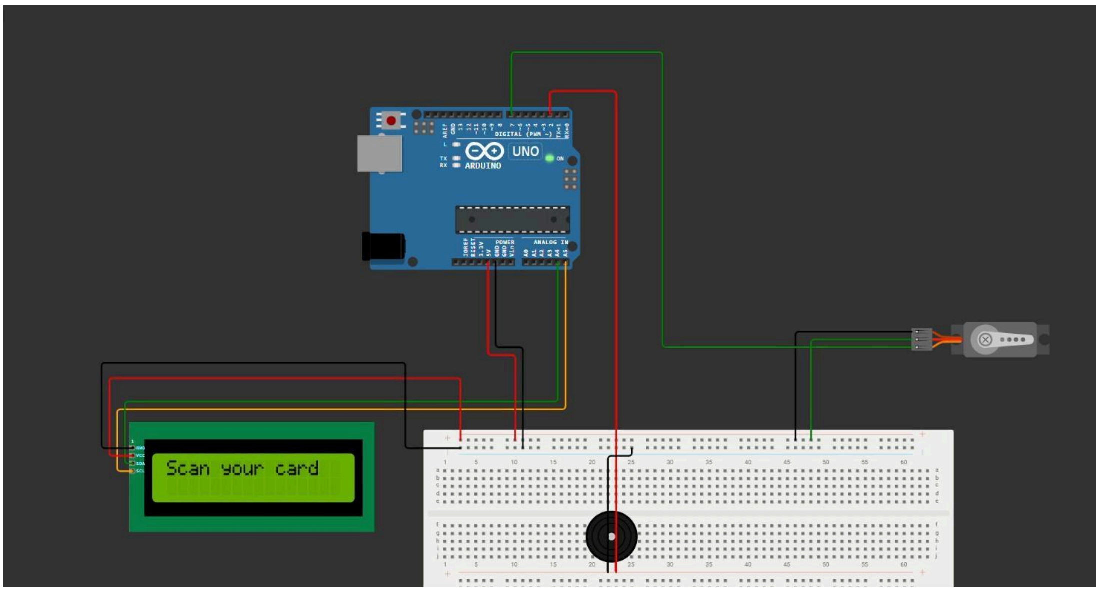
```
  Serial.begin(115200); // Start the serial communication

  Wire.begin(); // Start the I2C communication
  mpu.initialize(); // Initialize the MPU-6050 sensor

  // Check if the sensor is connected properly
  if (mpu.testConnection()) {
    Serial.println("MPU6050 connected successfully");
  } else {
    Serial.println("MPU6050 connection failed");
  }
}

void loop() {
  // Read accelerometer and gyroscope data
  int16_t ax, ay, az;
  int16_t gx, gy, gz;

  mpu.getAcceleration(&ax, &ay, &az); // Get accelerometer values
  mpu.getRotation(&gx, &gy, &gz); // Get gyroscope values

  // Print values to serial monitor
  Serial.print("Accelerometer: ");
  Serial.print("X = "); Serial.print(ax);
  Serial.print(", Y = "); Serial.print(ay);
  Serial.print(", Z = "); Serial.println(az);

  Serial.print("Gyroscope: ");
  Serial.print("X = "); Serial.print(gx);
  Serial.print(", Y = "); Serial.print(gy);
  Serial.print(", Z = "); Serial.println(gz);

  delay(500); // Delay for half a second
}
```

Practical Steps:

1. Setting Up the Project in Wokwi:
   - Go to www.wokwi.com and create a new project.
   - Choose the Arduino board you wish to simulate (e.g., Arduino Uno).
   - Add components to your project, including an LCD display, servo motor, buzzer, and a simulated RFID reader.
2. Writing and Uploading the Code:
   - Write the code as shown above.
   - Upload the code to your virtual Arduino board in Wokwi.

Simulation:





Conclusion:

This project demonstrates how to use Wokwi for simulating an RFID-based access control system with a servo barrier and LCD display. The simulation allows for testing and debugging without physical hardware, making it an efficient tool for prototyping and educational purpose